

PART-AName of the Student: E. Prem SathvikSignature of the Student: Roll No.: 230123022Signature of the Invigilator: **INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	1	9	15

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

$$\text{let } \beta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\text{LSE} \rightarrow Q(\beta) = \min \Rightarrow Q'(\beta) = 0$$

$$Q(\beta) = \sum (y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2$$

$$= (Y - X\beta)^T (Y - X\beta)$$

$$= Y^T Y - 2Y^T X\beta + \beta^T X^T X \beta$$

$$\frac{\partial Q'(\beta)}{\partial \beta} = 0 - 2X^T Y + 2X^T X(\beta)$$

$$\Rightarrow X^T Y = X^T X(\beta)$$

$$\beta = (X^T X)^{-1} X^T Y \quad \text{if } X^T X \text{ is invertible}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{45} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \frac{1}{45} \begin{bmatrix} 10 & 5 \\ 5 & 10 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y = \frac{1}{45} \begin{bmatrix} 10 & 5 \\ 5 & 10 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\frac{\partial}{\partial x} a x = a^T$$

$$\frac{\partial}{\partial x} a^T a x = a^T (a + a^T) x$$

$= 2a x$
if a is symmetric

$$B = \frac{1}{9} \times \begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}$$

$$= \frac{1}{3} \times \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}$$

$$= \frac{1}{3} \times \begin{bmatrix} Y_1 + Y_3 \\ Y_1 - Y_2 \end{bmatrix}$$

$$\therefore Q_1 = \frac{Y_1 + Y_3}{3}$$

$$Q_2 = \frac{Y_1 - Y_2}{3}$$

are the LSE of given regression problem.

(05)

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$a) \quad \mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \quad X_1 = U_1 \quad X_2 = U_2 \quad X_3 = U_1 + U_2$$

$$\rho_{12} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)} \sqrt{\text{Var}(X_2)}}$$

$$\text{Var}(X_1) = \text{Var}(U_1)$$

$$\text{Var}(X_2) = \text{Var}(U_2)$$

$$\text{Var}(X_3) = \text{Var}(U_1 + U_2)$$

$$\text{Cov}(U_1, U_2) = 0 \quad \because \text{They are independent}$$

$$\therefore \rho_{12} = \rho_{21} = 0$$

$$\rho_{11} = \rho_{22} = \rho_{33} = 1$$

$$\text{let } \text{Var}(U_1) = \sigma_1^2$$

$$\text{Var}(U_2) = \sigma_2^2$$

$$\begin{aligned} \text{Var}(X_3) &= \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2) \\ &= \sigma_1^2 + \sigma_2^2 + 0 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_1, X_3) &= \text{Cov}(U_1, U_1 + U_2) \\ &= E[(U_1 - \bar{U}_1)^2 + (U_1 - \bar{U}_1)(U_2 - \bar{U}_2)] \\ &= \sigma_1^2 \end{aligned}$$

$$\therefore \rho_{13} = \rho_{31} = \frac{\sigma_1^2}{\sqrt{\sigma_1^2} \sqrt{\sigma_1^2 + \sigma_2^2}} = \frac{\sigma_1}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

$$\text{Similarly } \rho_{32} = \rho_{23} = \frac{\sigma_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

$$\rho = \begin{bmatrix} 1 & 0 & \frac{\sigma_1}{\sqrt{\sigma_1^2 + \sigma_2^2}} \\ 0 & 1 & \frac{\sigma_2}{\sqrt{\sigma_1^2 + \sigma_2^2}} \\ \frac{\sigma_1}{\sqrt{\sigma_1^2 + \sigma_2^2}} & \frac{\sigma_2}{\sqrt{\sigma_1^2 + \sigma_2^2}} & 1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & \sigma_1^2 \\ 0 & \sigma_2^2 & \sigma_2^2 \\ \sigma_1^2 & \sigma_2^2 & \sigma_1^2 + \sigma_2^2 \end{bmatrix}$$

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$$b) \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 & \sigma_1^2 \\ 0 & \sigma_2^2 & \sigma_2^2 \\ \sigma_1^2 & \sigma_2^2 & \sigma_1^2 + \sigma_2^2 \end{bmatrix}$$

Eigenvectors of $\Sigma \quad \Rightarrow |\Sigma - \lambda I| = 0$

$$\therefore \begin{bmatrix} \sigma_1^2 - \lambda & 0 & \sigma_1^2 \\ 0 & \sigma_2^2 - \lambda & \sigma_2^2 \\ \sigma_1^2 & \sigma_2^2 & \sigma_1^2 + \sigma_2^2 - \lambda \end{bmatrix} = A$$

$$\det(A) = 0$$

$$(\sigma_1^2 - \lambda) [(\sigma_2^2 - \lambda)(\sigma_1^2 + \sigma_2^2 - \lambda) - \sigma_2^4] + \sigma_1^2 [(\lambda - \sigma_2^2)\sigma_1^2] = 0$$

$$(\sigma_1^2 - \lambda) [\sigma_1^2 \sigma_2^2 + \cancel{\sigma_1^4} - \lambda \sigma_2^2 - \lambda \sigma_1^2 - \lambda \sigma_2^2 + \lambda^2 - \cancel{\sigma_2^4}] + \lambda \sigma_1^4 - \sigma_1^4 \sigma_2^2 = 0$$

$$\cancel{\sigma_1^4 \sigma_2^2} - \lambda \sigma_1^2 \sigma_2^2 - \lambda \sigma_1^2 \sigma_2^2 + \lambda^2 \sigma_2^2 - \cancel{\lambda \sigma_1^4} + \lambda \sigma_1^2 - \lambda \sigma_1^2 \sigma_2^2 + \lambda^2 \sigma_2^2 + \lambda^2 \sigma_1^2 - \lambda^3$$

$$+ \lambda \cancel{\sigma_1^4} - \sigma_1^4 \cancel{\sigma_2^2} = 0$$

$$- 3\lambda \sigma_1^2 \sigma_2^2 + 2\lambda^2 \sigma_2^2 + 2\lambda^2 \sigma_1^2 - \lambda^3 = 0$$

$$\Rightarrow \lambda^3 - 2(\sigma_1^2 + \sigma_2^2)\lambda + 3\lambda \sigma_1^2 \sigma_2^2 = 0$$

$$\lambda (\lambda^2 - 2(\sigma_1^2 + \sigma_2^2)\lambda + 3\sigma_1^2 \sigma_2^2) = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = \lambda_3 = \lambda$$

Var of first PCA = $\lambda_{\max}$

$$= \frac{2(\sigma_1^2 + \sigma_2^2) + \sqrt{4\sigma_1^4 + 4\sigma_2^4 - 12\sigma_1^2 \sigma_2^2}}{2}$$

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

i. Complete linkage hierarchical procedure.

ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) (i)

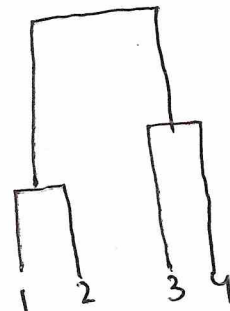
	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

Complete linkage. $d_{uw} = \max \{ d_{ij}, i \in u, j \in w \}$

$d(12), 3$
 $= \max(d_{12}, d_{13}, d_{23})$
 $= 11$

$d(12), 4$
 $= \max(d_{14}, d_{24})$
 $= 5$

	(12)	3	4
(12)	0		
3	11	0	
4	5	3.5	0



$d((12), 34) = \max(d(12), 3, d(12), 4)$

(ii)

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

$d_{uw} = \frac{\sum_i \sum_j d_{ij}}{N_u N_w}$

$d(12), 3$
 $= \frac{d_{13} + d_{23}}{2}$

	(12)	3	4
(12)	0		
3	6.5	0	
4	4	3.5	0

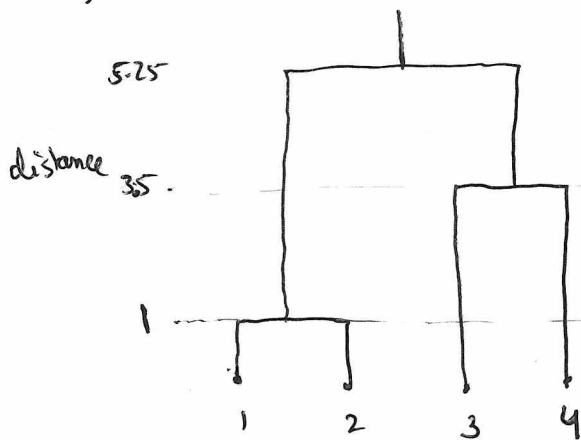
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	(12)	(34)
(12)	0	
(34)	5.25	0

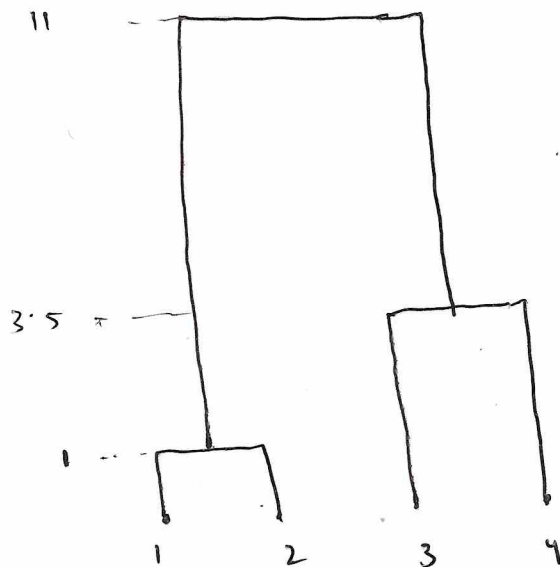
$d(12), 34$
 $= \frac{d_{13} + d_{14} + d_{23} + d_{24}}{2 \times 2}$
 $= \frac{11 + 5 + 2 + 3}{4}$
 $= \frac{21}{4} = 5.25$

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b)



Average linkage
~~Complete~~ hierarchical procedure.



Complete linkage
~~A~~ hierarchical procedure.

Both procedures give the same steps in this example but give different distances for each group because.

Complete linkage calculates the maximum distance possible between the two points of the two groups $\therefore$ Complete linkage distances are higher

In Avg linkage procedure Everytime we calculate the diff between two clusters we calculate the Avg value of the distances b/w each points of the two clusters.

ADDITIONAL PAGES FOR ANSWERS
